Homework solutions for 02YELMA
Chapter: Electromagnetic waves

1. The plane wave propagating in the z-direction is

$$\mathbf{E}(z,t) = \mathbf{E}_0\, e^{i(kz-\omega t)}, \qquad \mathbf{E}_0 = \left[(\mathrm{E}_0)_x\, \hat{x} + (\mathrm{E}_0)_y\, \hat{y} + (\mathrm{E}_0)_z\, \hat{z}\right].$$

Since $\mathbf{E}_0$ is a constant vector and the spatial dependence is only through z, we compute

$$\vec{\nabla}\cdot\mathbf{E} = \frac{\partial \mathrm{E}_x}{\partial x} + \frac{\partial \mathrm{E}_y}{\partial y} + \frac{\partial \mathrm{E}_z}{\partial z}.$$

The first two terms vanish because E_x and E_y do not depend on x or y, respectively. The third term gives

$$\frac{\partial \mathrm{E}_z}{\partial z} = (\mathrm{E}_0)_z\,(ik)\,e^{i(kz-\omega t)}.$$

Applying Gauss's law in vacuum, $\vec{\nabla}\cdot\vec{E} = 0$:

$$(\mathrm{E}_0)_z\,(ik)\,e^{i(kz-\omega t)} = 0.$$

Since $k \neq 0$ and $e^{i(kz-\omega t)} \neq 0$ for all z, t, we conclude

$$\boxed{(E_0)_z = 0.}$$

The electric field has no component along the direction of propagation; it is purely transverse.

Yes, an identical argument applies to the magnetic field. From $\vec{\nabla}\cdot\vec{B} = 0$ and $\mathbf{B} = \mathbf{B}_0\, e^{i(kz-\omega t)}$, the same steps yield $(B_0)_z = 0$. Electromagnetic waves in vacuum are **transverse** in both $\vec{E}$ and $\vec{B}$.

2. Since all fields depend only on z and t, and $(E_0)_z = 0$:

$$\vec{\nabla}\times\mathbf{E} = -\frac{\partial \mathrm{E}_y}{\partial z}\,\hat{x} + \frac{\partial \mathrm{E}_x}{\partial z}\,\hat{y} = \left[-ik\,(\mathrm{E}_0)_y\,\hat{x} + ik\,(\mathrm{E}_0)_x\,\hat{y}\right]e^{i(kz-\omega t)}.$$

Time derivative of the magnetic field.

$$-\frac{\partial \mathbf{B}}{\partial t} = i\omega\,\mathbf{B}_0\,e^{i(kz-\omega t)}.$$

Setting Faraday's law $\vec{\nabla}\times\mathbf{E} = -\partial\mathbf{B}/\partial t$ and comparing components:

$$\hat{x}: \quad -ik\,(\mathrm{E}_0)_y = i\omega\,(\mathrm{B}_0)_x \implies \boxed{(\mathrm{B}_0)_x\,\frac{\omega}{k} = -(\mathrm{E}_0)_y}$$

$$\hat{y}: \quad ik\,(\mathrm{E}_0)_x = i\omega\,(\mathrm{B}_0)_y \implies \boxed{(\mathrm{B}_0)_y\,\frac{\omega}{k} = (\mathrm{E}_0)_x}$$

Compact vector form. Compute the cross product $\hat{z} \times \mathbf{E}_0$ using $(E_0)_z = 0$:

$$\hat{z} \times \mathbf{E}_0 = (\mathrm{E}_0)_x\,(\hat{z} \times \hat{x}) + (\mathrm{E}_0)_y\,(\hat{z} \times \hat{y}) = (\mathrm{E}_0)_x\,\hat{y} - (\mathrm{E}_0)_y\,\hat{x}.$$

Multiplying by k/ω:

$$\frac{k}{\omega}\,(\hat{z} \times \mathbf{E}_0) = -\frac{k}{\omega}(\mathrm{E}_0)_y\,\hat{x} + \frac{k}{\omega}(\mathrm{E}_0)_x\,\hat{y}.$$

This reproduces exactly the two component relations above, and the z-component is automatically zero. Combined with $(E_0)_z = (B_0)_z = 0$, all four conditions are captured by the single equation:

$$\boxed{\mathbf{B}_0 = \frac{k}{\omega}\,(\hat{z} \times \mathbf{E}_0)\,.}$$

Physical interpretation. In vacuum:

- $\vec{E}$ and $\vec{B}$ are both **transverse** (perpendicular to the propagation direction $\hat{z}$).
- $\vec{E}$ and $\vec{B}$ are **mutually perpendicular** (the cross product $\hat{z} \times \vec{E}$ rotates $\vec{E}$ by 90° in the transverse plane).
- Their amplitudes are related by $B_0 = E_0/c$, since $\omega/k = c$ in vacuum.
- They oscillate **in phase**.

3. Given: $\vec{E}(z,t) = E_0 \cos(kz - \omega t)\,\hat{x}$.

(a) Magnetic field

Using the result $\vec{B} = \frac{k}{\omega}(\hat{z} \times \vec{E}) = \frac{1}{c}(\hat{z} \times \vec{E})$:

$$\vec{B}(z,t) = \frac{E_0}{c}\cos(kz - \omega t)\,(\hat{z} \times \hat{x}) = \boxed{\frac{E_0}{c}\cos(kz - \omega t)\,\hat{y}.}$$

(b) Instantaneous Poynting vector

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B} = \frac{1}{\mu_0}\,[E_0 \cos(kz - \omega t)\,\hat{x}] \times \left[\frac{E_0}{c}\cos(kz - \omega t)\,\hat{y}\right]$$

$$\boxed{\vec{S} = \frac{E_0^2}{\mu_0 c}\cos^2(kz - \omega t)\,\hat{z}.}$$

Since $\cos^2 \geq 0$, $\mu_0 > 0$, and $c > 0$, we have $\vec{S} \parallel +\hat{z}$, confirming that **energy flows in the direction of propagation**.

(c) Time-averaged Poynting vector

Since $\langle \cos^2(kz - \omega t) \rangle = \frac{1}{2}$:

$$\boxed{\langle \vec{S} \rangle = \frac{E_0^2}{2\mu_0 c}\,\hat{z}.}$$

(d) Electromagnetic energy density

$$u = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2\mu_0}B^2 = \frac{1}{2}\varepsilon_0 E_0^2 \cos^2(kz - \omega t) + \frac{1}{2\mu_0}\frac{E_0^2}{c^2}\cos^2(kz - \omega t).$$

Using $c^2 = 1/(\mu_0\varepsilon_0)$, so $1/(\mu_0 c^2) = \varepsilon_0$:

$$u = \underbrace{\frac{1}{2}\varepsilon_0 E_0^2 \cos^2(kz - \omega t)}_{u_E} + \underbrace{\frac{1}{2}\varepsilon_0 E_0^2 \cos^2(kz - \omega t)}_{u_B}$$

$$\boxed{u = \varepsilon_0 E_0^2 \cos^2(kz - \omega t).}$$

We see $u_E = u_B$ term by term: **the electric and magnetic fields contribute equally to the total energy density.** The time average is $\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2$.

4. A parallel-plate capacitor with circular plates of radius R, separation $d \ll R$, is charged by a steady current I. Between the plates: $\vec{E}(t) = E(t)\,\hat{z}$, uniform.

(a) Magnetic field between the plates

There is no free current between the plates. Maxwell–Ampère's law gives

$$\oint \vec{B} \cdot d\vec{l} = \mu_0\varepsilon_0 \frac{\partial}{\partial t} \int \vec{E} \cdot d\vec{A}.$$

By symmetry, $\vec{B} = B(r)\,\hat{\phi}$. For a circular Ampèrian loop of radius $r < R$:

$$B(r)\,(2\pi r) = \mu_0\varepsilon_0\,\dot{E}\,(\pi r^2).$$

Since the total displacement current through the full cross-section equals the conduction current, $\varepsilon_0\,\dot{E}\,\pi R^2 = I$, so $\varepsilon_0\,\dot{E} = I/(\pi R^2)$. Substituting:

$$\boxed{\vec{B}(r) = \frac{\mu_0 I\, r}{2\pi R^2}\,\hat{\phi}, \qquad r < R.}$$

(b) Poynting vector

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B} = \frac{1}{\mu_0}\,E(t)\,\hat{z} \times \frac{\mu_0 I\, r}{2\pi R^2}\,\hat{\phi} = \frac{E(t)\, I\, r}{2\pi R^2}\,(\hat{z} \times \hat{\phi}).$$

Using the cylindrical coordinate identity $\hat{z} \times \hat{\phi} = -\hat{r}$:

$$\boxed{\vec{S} = -\frac{E(t)\, I\, r}{2\pi R^2}\,\hat{r}.}$$

The Poynting vector points **radially inward** (in the $-\hat{r}$ direction).

(c) Energy flows in from the sides

The Poynting vector has **no** $\hat{z}$-component anywhere between the plates—it is purely radial. Therefore, no energy enters through the top or bottom surfaces (the plates). All electromagnetic energy enters the capacitor volume through the **cylindrical side surface** at $r = R$.

(d) Total rate of energy flow

Integrate $\vec{S}$ over the cylindrical side surface at $r = R$ (outward normal $\hat{n} = +\hat{r}$, area element $dA = R\,d\phi\,dz$):

$$P_{\text{in}} = -\oint_{\text{side}} \vec{S} \cdot \hat{r}\, dA = -\int_0^d \int_0^{2\pi} \left(-\frac{E(t)\, I}{2\pi} \right) d\phi\, dz$$

$$\boxed{P_{\text{in}} = E(t)\, I\, d.}$$

(e) Agreement with stored energy

The electrostatic energy stored between the plates is

$$U = \frac{1}{2}\varepsilon_0 E^2\, \pi R^2 d.$$

Its rate of change:

$$\frac{dU}{dt} = \varepsilon_0\, \pi R^2\, d\, E\, \dot{E} = \varepsilon_0\, \pi R^2\, d\, E\, \frac{I}{\varepsilon_0 \pi R^2} = E\, I\, d.$$

$$\boxed{\frac{dU}{dt} = E(t)\, I\, d = P_{\text{in}}.} \quad \checkmark$$

The rate at which field energy increases inside the capacitor exactly equals the electromagnetic energy flux entering through the sides.

(f) Conceptual significance

This example illustrates three key ideas:

(a) **Displacement current is real:** Even though no charges flow between the plates, the time-varying electric field ($\varepsilon_0\, \partial \vec{E}/\partial t$) acts as a source of $\vec{B}$, exactly as Maxwell postulated. Without it, Ampère's law would be inconsistent.

(b) **The Poynting vector tracks energy flow:** Energy does not travel down the wires and jump across the gap. Instead, $\vec{S}$ shows that energy is stored in the electromagnetic field and flows laterally into the capacitor volume from the surrounding space.

(c) **Energy conservation via Poynting's theorem:** The equality $dU/dt = P_{\text{in}}$ is a direct verification of Poynting's theorem, $\partial u/\partial t = -\vec{\nabla} \cdot \vec{S}$, confirming the self-consistency of Maxwell's equations and the physical reality of field energy transport.

5. Given:

$$\vec{E}(z,t) = \hat{x}\, E_0 \cos(kz - \omega t) + \hat{y}\, E_0 \cos(kz - \omega t + \phi).$$

(a) Trajectory in the xy-plane

At a fixed position $z = z_0$, define $\theta \equiv kz_0 - \omega t$. Then

$$E_x = E_0 \cos\theta, \qquad E_y = E_0 \cos(\theta + \phi).$$

Expanding: $E_y = E_0[\cos\theta \cos\phi - \sin\theta \sin\phi]$, so

$$E_y - E_x \cos\phi = -E_0 \sin\theta \sin\phi.$$

From $E_x = E_0 \cos\theta$ we get $E_0^2 \sin^2\theta = E_0^2 - E_x^2$. Squaring both sides of the equation above:

$$(E_y - E_x \cos\phi)^2 = (E_0^2 - E_x^2) \sin^2\phi.$$

Expanding and rearranging:

$$\boxed{\frac{E_x^2}{E_0^2} + \frac{E_y^2}{E_0^2} - \frac{2E_xE_y}{E_0^2}\cos\phi = \sin^2\phi.}$$

This is the equation of an **ellipse** in the E_x–E_y plane (the polarization ellipse), which degenerates into a line or circle for special values of ϕ.

(b) Special polarization states

Linear polarization occurs when the ellipse degenerates into a straight line, i.e. $\sin^2\phi = 0$:

$$\boxed{\phi = 0 \quad \text{or} \quad \phi = \pi \quad (\text{more generally } \phi = n\pi,\ n \in \mathbb{Z}).}$$

- $\phi = 0$: $(E_x - E_y)^2 = 0 \;\Rightarrow\; E_y = E_x$, a line at 45°.
- $\phi = \pi$: $(E_x + E_y)^2 = 0 \;\Rightarrow\; E_y = -E_x$, a line at $-45°$.

Circular polarization occurs when the ellipse becomes a circle, requiring $\cos\phi = 0$ (so the cross term vanishes) with equal amplitudes in both components:

$$\boxed{\phi = +\frac{\pi}{2} \quad \text{or} \quad \phi = -\frac{\pi}{2}.}$$

The trajectory becomes $E_x^2 + E_y^2 = E_0^2$, a circle of radius E_0.

(c) Sense of rotation

Set $z_0 = 0$ so $\theta = -\omega t$.

Case $\phi = +\pi/2$:

$$E_x = E_0 \cos(\omega t), \qquad E_y = -E_0 \sin(\omega t).$$

ωt	E_x	E_y
0	E_0	0
$\pi/2$	0	$-E_0$

The tip rotates $\hat{x} \to -\hat{y}$: **clockwise** when viewed looking in the $+z$-direction (right-circular polarization).

Case $\phi = -\pi/2$:

$$E_x = E_0 \cos(\omega t), \qquad E_y = E_0 \sin(\omega t).$$

The tip rotates $\hat{x} \to +\hat{y}$: **counterclockwise** when viewed looking in the $+z$-direction (left-circular polarization).

(d) Time-averaged Poynting vector

The magnetic field follows from $\vec{B} = \frac{1}{c}(\hat{z} \times \vec{E})$:

$$\vec{B} = \frac{E_0}{c}\left[-\cos(kz - \omega t + \phi)\,\hat{x} + \cos(kz - \omega t)\,\hat{y}\right].$$

The Poynting vector is

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}.$$

Using the vector identity (with $E_z = 0$):

$$\vec{E} \times (\hat{z} \times \vec{E}) = (\vec{E}\cdot\vec{E})\,\hat{z} - (\vec{E}\cdot\hat{z})\,\vec{E} = |\vec{E}|^2\,\hat{z},$$

we obtain

$$\vec{S} = \frac{|\vec{E}|^2}{\mu_0 c}\,\hat{z}.$$

Now compute the time average of $|\vec{E}|^2$:

$$|\vec{E}|^2 = E_0^2 \cos^2(kz - \omega t) + E_0^2 \cos^2(kz - \omega t + \phi).$$

Since $\langle \cos^2(\cdot) \rangle = 1/2$ regardless of any constant phase offset:

$$\langle |\vec{E}|^2 \rangle = \frac{E_0^2}{2} + \frac{E_0^2}{2} = E_0^2.$$

Therefore,

$$\boxed{\langle \vec{S} \rangle = \frac{E_0^2}{\mu_0 c}\,\hat{z}.}$$

This points along $+\hat{z}$, the direction of propagation, and **its magnitude is independent of ϕ.**

Why is $\langle \vec{S} \rangle$ independent of ϕ? The phase ϕ redistributes energy between the two transverse components over the cycle—it changes the *shape* of the polarization ellipse—but it does not change the *amplitude* of either component. Each component independently contributes $E_0^2/2$ to the time-averaged intensity. Since the total time-averaged power depends only on $\langle E_x^2 \rangle + \langle E_y^2 \rangle$, and cross terms average to zero, the result is the same for linear, circular, or any elliptical polarization sharing the same component amplitudes.